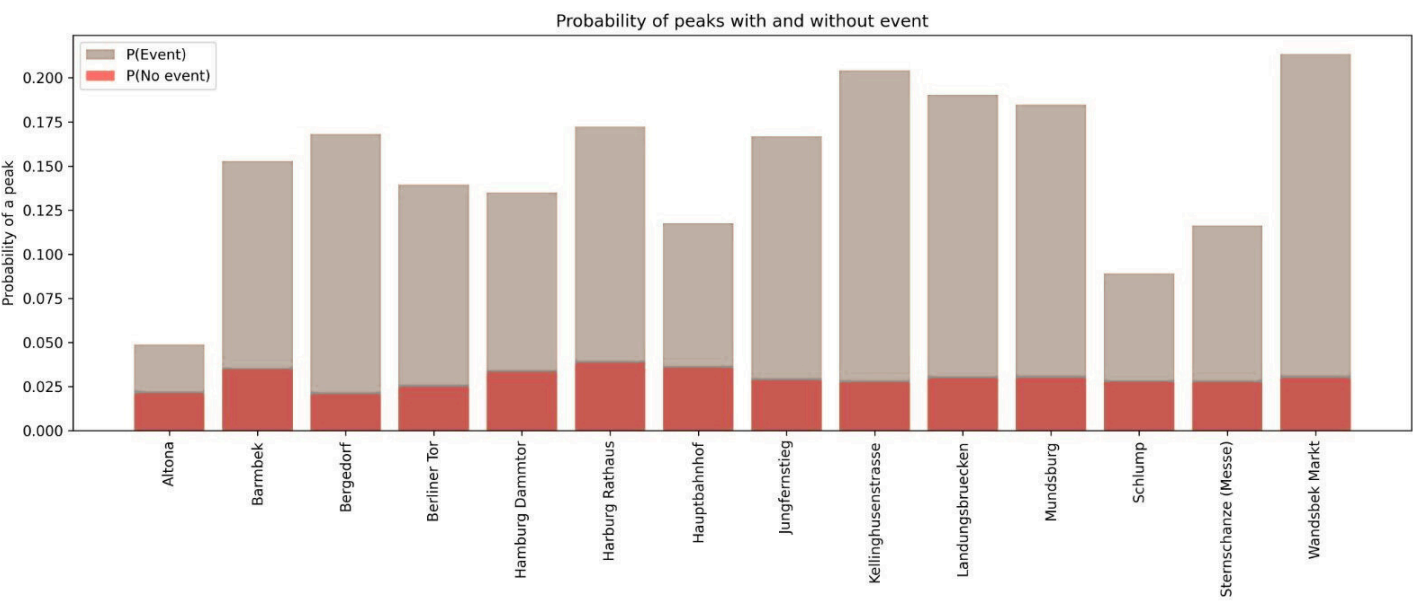
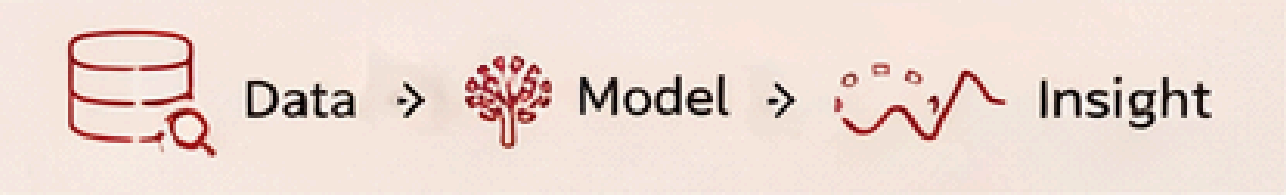


How Hamburg Moves

Data meets motion – Visualizing mobility, events, and behavior in Hamburg

From Numbers to Movement

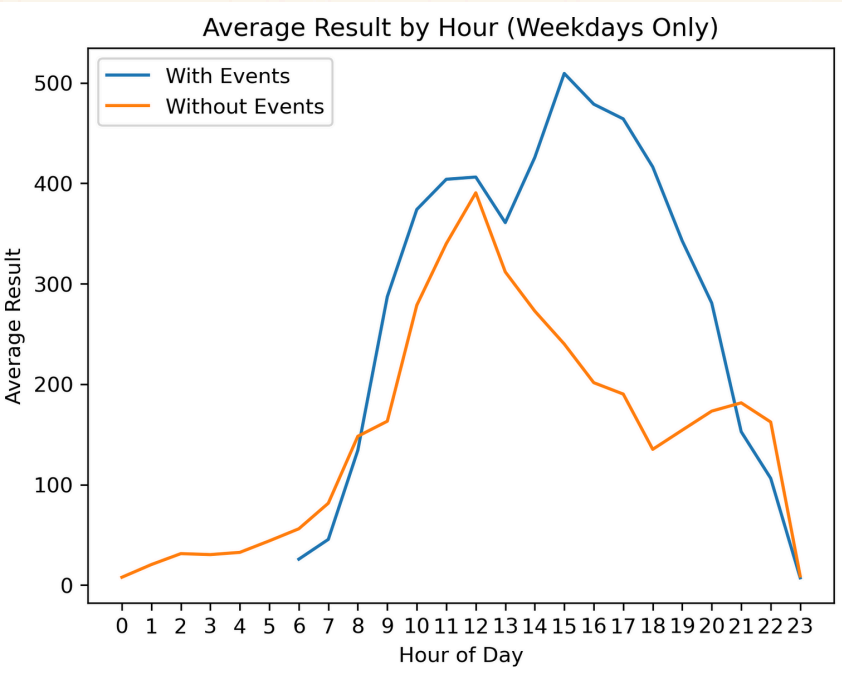
After exploring various datasets, we chose to study how Hamburg moves – how commuters, tourists, and events shape urban rhythm. Using HVV data and machine-learning transport data into stories of flow, delay, and emotion.



From Numbers to Movement

AI predicts the crowd – and almost gets it perfect

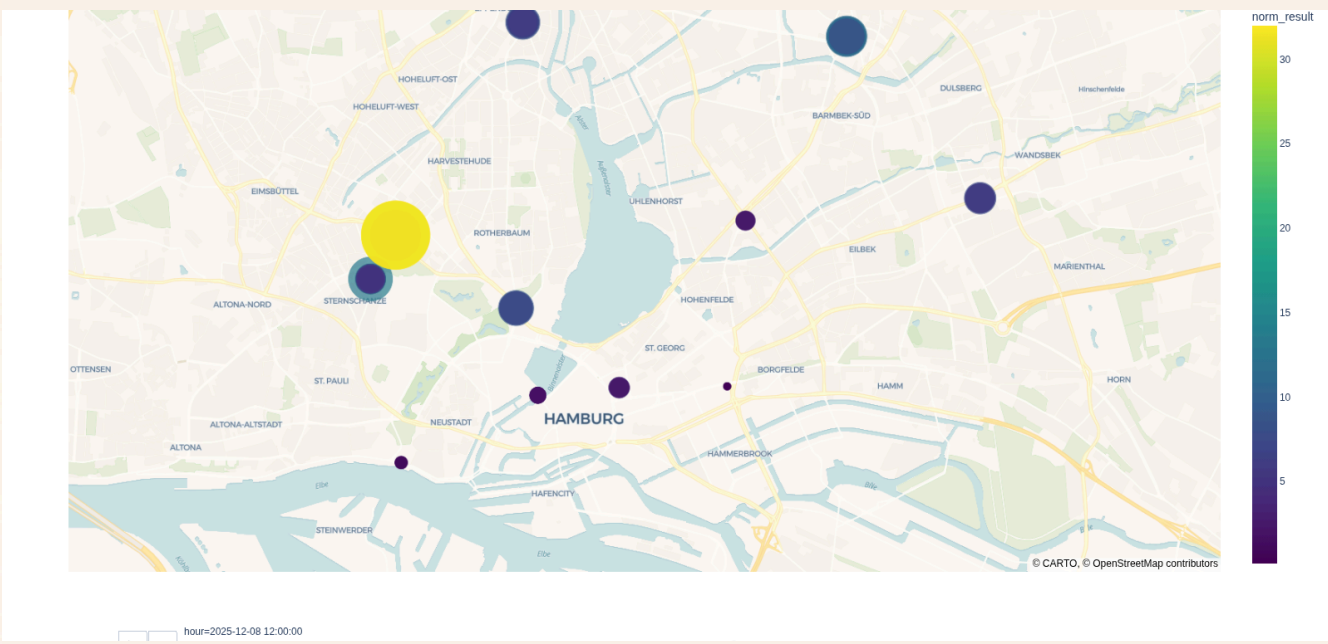
Machine learning predicts passenger flows at Altona station during the holidays. Algorithms like XGBoost and Random Forest can predict your routine but not your spontaneous outings



When Events Set the City in Motion

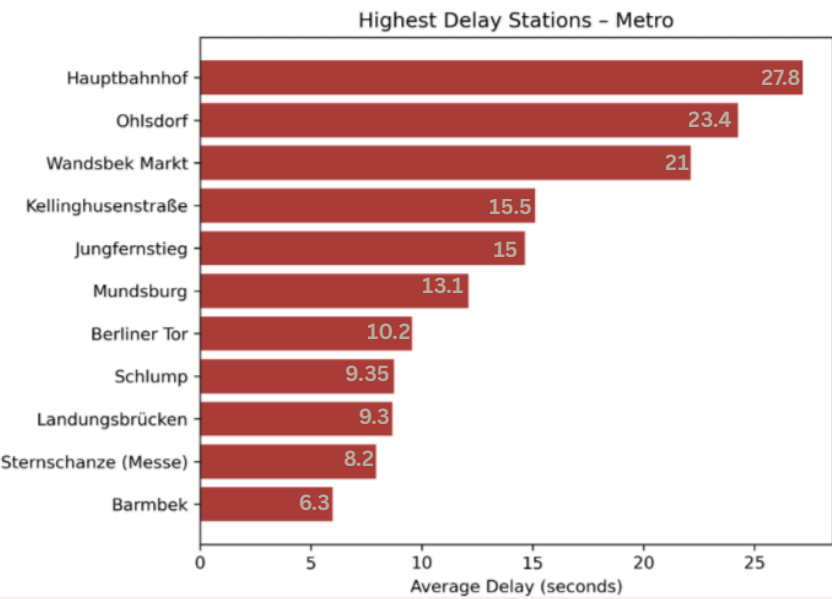
Event-driven peaks dominate – especially at Wandsbek Markt, Kellinghusenstraße, and Bergedorf – where up to one in four movement surges align with major city happenings.

Hamburg’s core glows yellow; the suburbs breathe slower

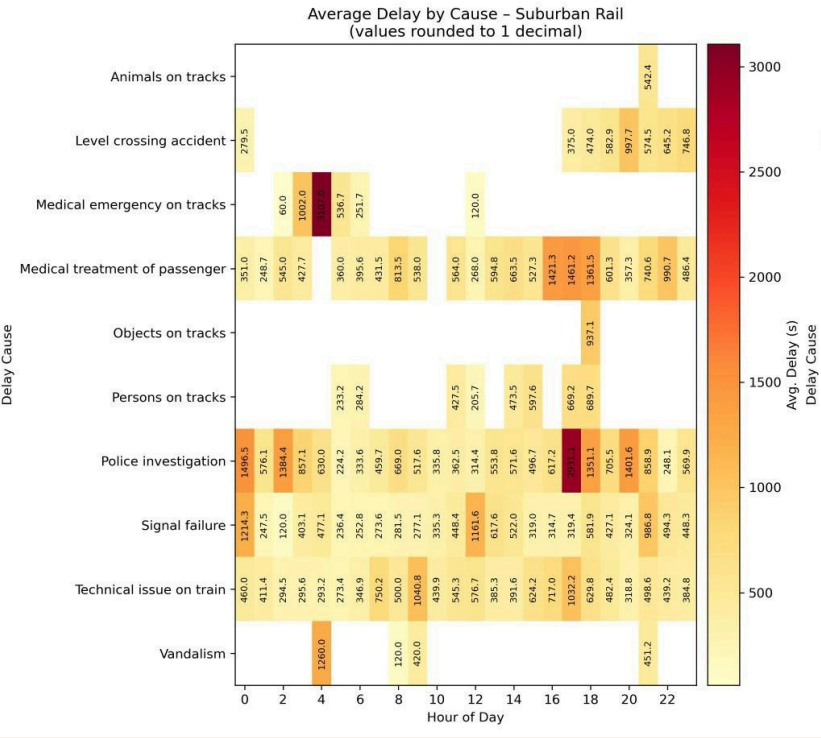


A heatmap of movement at 12 PM, December 8. Jungfernstieg, Sternschanze, and Hauptbahnhof pulse with life; cooler suburban areas show a gentler rhythm.

Where time stands still – Hauptbahnhof leads the list



Hauptbahnhof tops the delay charts. Regional trains suffer the longest wait times; metros face frequent but shorter slowdowns.



Why Trains Run Late

The heatmaps reveal when and why delays occur across Hamburg’s rail systems. Morning peaks are dominated by medical and police incidents, while afternoon slowdowns stem from signal and technical faults.

What the City Taught Us

We found that cities have memory. Their movement favors – but never forgets – daily events. Short-term forecasts are reliable, but human spontaneity remains the real variable. Understanding this rhythm makes mobility planning human, not mechanical.

